

Multiplexed Networks for Embedded Systems

CAN, LIN, Flexray, Safe-by-Wire...

Dominique Paret

Translated by

Roderick Riesco, MA

Member of the Institute of Translation and Interpreting, UK



John Wiley & Sons, Ltd

Multiplexed Networks for Embedded Systems

Multiplexed Networks for Embedded Systems

CAN, LIN, Flexray, Safe-by-Wire...

Dominique Paret

Translated by

Roderick Riesco, MA

Member of the Institute of Translation and Interpreting, UK



John Wiley & Sons, Ltd

Originally published in the French language by Dunod as “Dominique Paret: Réseaux multiplexes pour systèmes embarqués. CAN, LAN, FlexyRay, Safe-by-Wire”. Copyright © Dunod, Paris 2005.

Copyright © 2007 John Wiley & Sons Ltd, The Atrium, Southern Gate, Chichester,
West Sussex PO19 8SQ, England
Telephone (+44) 1243 779777

Email (for orders and customer service enquiries): cs-books@wiley.co.uk

Visit our Home Page on www.wiley.com

All Rights Reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning or otherwise, except under the terms of the Copyright, Designs and Patents Act 1988 or under the terms of a licence issued by the Copyright Licensing Agency Ltd, 90 Tottenham Court Road, London W1T 4LP, UK, without the permission in writing of the Publisher. Requests to the Publisher should be addressed to the Permissions Department, John Wiley & Sons Ltd, The Atrium, Southern Gate, Chichester, West Sussex PO19 8SQ, England, or emailed to permreq@wiley.co.uk, or faxed to (+44) 1243 770620.

Designations used by companies to distinguish their products are often claimed as trademarks. All brand names and product names used in this book are trade names, service marks, trademarks or registered trademarks of their respective owners. The Publisher is not associated with any product or vendor mentioned in this book.

This publication is designed to provide accurate and authoritative information in regard to the subject matter covered. It is sold on the understanding that the Publisher is not engaged in rendering professional services. If professional advice or other expert assistance is required, the services of a competent professional should be sought.

Other Wiley Editorial Offices

John Wiley & Sons Inc., 111 River Street, Hoboken, NJ 07030, USA

Jossey-Bass, 989 Market Street, San Francisco, CA 94103-1741, USA

Wiley-VCH Verlag GmbH, Boschstr. 12, D-69469 Weinheim, Germany

John Wiley & Sons Australia Ltd, 42 McDougall Street, Milton, Queensland 4064, Australia

John Wiley & Sons (Asia) Pte Ltd, 2 Clementi Loop #02-01, Jin Xing Distripark, Singapore 129809

John Wiley & Sons Canada Ltd, 6045 Freemont Blvd, Mississauga, ONT, Canada L5R 4J3

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic books.

Anniversary Logo Design: Richard J. Pacifico

British Library Cataloguing in Publication Data

A catalogue record for this book is available from the British Library

ISBN 978-0-470-03416-3 (HB)

Typeset in 10/12 pt Times by Thomson Digital

Printed and bound in Great Britain by Antony Rowe Ltd, Chippenham, Wiltshire

This book is printed on acid-free paper responsibly manufactured from sustainable forestry in which at least two trees are planted for each one used for paper production.

I dedicate this book to my father, André Paret, who passed away many years ago, and who, wherever he is now, will surely be smiling wickedly to himself when he finds that X-by-Wire solutions are about to be introduced in the car industry in a few years' time. He was a pioneer in mechanical and electronic systems for civil aviation, and, more than fifty years ago, brought such expressions as 'fly-by-wire', 'level 5, 6 and 7 multiple redundancy', 'reliability', 'lambda', etc., into my childhood and teenage years.

Now the circle is unbroken. Life is always beginning again!

DOMINIQUE

Contents

Preface	xi
Acknowledgements	xv
Part A CAN: from concept to reality	1
1 The CAN bus: general	3
1.1 Concepts of bus access and arbitration	3
1.2 Error processing and management	8
1.3 Increase your word power	12
1.4 From concept to reality	12
1.5 Historical context of CAN	14
1.6 Patents, licences and certification	19
2 CAN: its protocol, its properties, its novel features	25
2.1 Definitions of the CAN protocol: ‘ISO 11898-1’	25
2.2 Errors: their intrinsic properties, detection and processing	54
2.3 The rest of the frame	71
2.4 CAN 2.0B	76
3 The CAN physical layer	83
3.1 Introduction	83
3.2 The ‘CAN bit’	86
3.3 Nominal bit time	90
3.4 CAN and signal propagation	94
3.5 Bit synchronization	106
3.6 Network speed	116
4 Medium, implementation and physical layers in CAN	125
4.1 The range of media and the types of coupling to the network	126
4.2 High speed CAN, from 125 kbit s ⁻¹ to 1 Mbit s ⁻¹ : ISO 11898-2	131
4.3 Low speed CAN, from 10 to 125 kbit s ⁻¹	142

4.4	Optical media	166
4.5	Electromagnetic media	169
4.6	Pollution and EMC conformity	169
5	Components, applications and tools for CAN	179
5.1	CAN components	179
5.2	Applications	199
5.3	Application layers and development tools for CAN	215
6	Time-triggered protocols – FlexRay	231
6.1	Some general remarks	231
6.2	Event-triggered and time-triggered aspects	232
6.3	TTTAN – Time-triggered communication on CAN	233
6.4	Towards high-speed, X-by-Wire and redundant systems	236
6.5	FlexRay	242
Part B	New multiplexed bus concepts: LIN, FlexRay, Fail-safe SBC, Safe-by-Wire	273
7	LIN – Local Interconnect Network	285
7.1	Introduction	285
7.2	Basic concept of the LIN 2.0 protocol	287
7.3	Cost and market	301
7.4	Conformity of LIN	301
7.5	Examples of components for LIN 2.0	302
8	Think ‘Bus’, think ‘Fail-safe SBC’, ‘Gateways’ ...	313
8.1	Fail-safe SBCs: their multiple aspects and reasons for using them	314
8.2	The strategy and principles of re-use	334
8.3	Demo board	335
8.4	Gateways	337
8.5	Managing the application layers	338
9	Safe-by-Wire	343
9.1	A little history	343
9.2	Safe-by-Wire Plus	345
9.3	Some words of technology	347
10	Audio-video buses	359
10.1	I2C Bus	359
10.2	The D2B (Domestic digital) bus	360
10.3	The MOST (Media oriented systems transport) bus	364
10.4	The IEEE 1394 bus or ‘FireWire’	371

11 RF communication and wireless mini-networks	377
11.1 Radio-frequency communication: internal	377
11.2 Radio-frequency communication: external	381
11.3 Wireless networks	394
Conclusion	399
Part C Appendices	401
Appendix A. CiA (CAN in Automation)	403
Appendix B. Essential references	407
Appendix C. Further reading	411
Appendix D. Useful addresses	413
Index	415

Preface

Dear reader,

Multiplexed networks such as CAN, LIN and other types are now a mature industrial sector worldwide, and only a few new systems like X-by-Wire are still awaiting the final touches.

Having worked in this field for many years, I felt the need to draw up a report on these newer topics. Very little basic and application-related information and technical training is currently available for engineers, technicians and students. We trust that this book will at least partially make up for this deficiency.

I have waited a long time before taking up my pen again (rather difficult, with a PC!) to write this new volume. This is because many novel products have appeared, accompanied by their inevitable publicity constantly proclaiming that ‘everything is good, everything is fine’. I decided to wait for a while until the dust had cleared and we could begin to see the horizon again; as usual, however, this took some considerable time.

In order not to fall into this trap of exaggerated publicity, let me clarify the situation straight away. At present, the version of LIN known as rev. 2.0 has been stable since the end of September 2003, Safe-by-Wire Plus rev. 2.0 since August 2004 and the final version of FlexRay, rev. 2.1, was officially released in March 2005. After these three births and our sincere congratulations to the happy parents, there is no reason to wait any longer before describing their characteristics.

The aim of this book is therefore to provide the most comprehensive guide at a given moment to this rapidly developing field. This book is not intended to be encyclopaedic, but aims to be a long and thorough technical introduction to this topic (and not just a literal translation of what anybody can find on the Web). It is thorough in the sense that all the ‘real’ aspects of these applications (principles, components, standards, applications, security, etc.) are dealt with in detail. For newcomers to this field, the book offers a global overview of the concepts and applications of the various buses.

I am aware that this sector is developing rapidly, and the content of this book will have to be updated in about three or four years. In the meantime, I can at least set out the basics and the fundamental principles.

I have also done my best to make this volume educational, enabling the reader to make the connection between theory, technology and economics, etc., at all times.

To complete this preface, I should inform you that another book by the same author is published under the title *Le bus CAN –Applications* [*The CAN bus – Applications*] which complements the present volume by dealing more specifically with the application layers of CAN and the details of their implementation; this should meet the requirements of the vast

majority of users. However, if you still have even the shadow of a doubt about any of these matters, you are always welcome to contact me by post or e-mail with your questions.

Meanwhile, I hope you gain both pleasure and benefit from this book; above all, enjoy it, because I wrote it not for myself but for you!

Very important note. I must immediately draw readers' attention to the important fact that, in order to cover the topic of multiplexed buses and networks in a comprehensive way, this book describes a large number of technical principles which are patented or subject to licensing and associated rights (bit coding, communication methods, etc.), and which have already been published in official professional technical communications or at public conferences, seminars, etc.

Any application of these principles must comply with the current law in all circumstances.

How to use this book

You are advised to read the next few lines carefully before moving on to the technical details which are provided in the following chapters.

First of all, you should be aware that this book covers many topics which impact on each other, merge and intersect, making it difficult to devise an overall plan. This led me to opt for an overall educational approach to enable readers to make sense of all these communication principles and emerging new protocols which they will discover and use in the future.

The introduction is designed to whet your appetite with a look at an everyday application, namely the mysteries of the on-board and In-Vehicle Network (IVN) systems of motor vehicles. Of course, everything described in this book is equally relevant to all kinds of industrial applications (control of machine tools and production lines, avionics, automation systems for large buildings, etc.).

Part A

Chapters 1–5, about CAN, are intended to bring you up to speed with the current state of the CAN protocol, all the possible subdivisions of the physical layers and everything relating to conformity problems. Some of you will already be aware of some of this information, but many complementary details of the physical layers have been added or updated. We cannot discuss these topics without (briefly) mentioning the CAL, CAN Open, OSEK and AUTOSAR application layers and the hardware and software tools needed to support assistance with development, checking, production, maintenance, etc.

In Chapter 6, I describe the limits of CAN in terms of functions and applications, and clarify the distinction between event-triggered and time-triggered communication systems, pointing out all the implications of what are known as real-time security applications. This leads to an introduction to the function and content of protocols such as TTCAN and FlexRay, as well as applications of the X-by-Wire type for the latter; in plain English, this means: good-bye to mechanical systems, from now on everything is 'by-wire'. A good start!

At this point we shall have completed the description of what are known as the high-level protocols, and we shall be able to start on the second major section of this book, concerned with other members of the CAN family and the possible relationships between all these buses.

Part B

I shall start with a detailed description of the new LIN bus (Chapter 7), its development, its properties, its problems and the way to overcome them. The LIN bus is generally presented by its designers as a sub-type of CAN, where the ‘sub’ is not pejorative, but purely functional. I shall go on to describe (in Chapter 8) the various constraints and possibilities for gateways between the buses described in Part One and these new arrivals, explaining how the Fail-safe – system basis chip (SBC) and other gateways are designed and constructed.

Finally, to complete this long ‘bus trip’, I shall describe (with only a few details, to avoid having to provide a universal encyclopaedia) the multiplicity of other buses surrounding those described in the previous two sections in on-board systems like those fitted in motor vehicles. Some experts consider that the motor car will eventually become a safe, reliable means of mobility, a domestic outpost (with audio, video, games, etc.), and an extension of the office! On this topic, I shall describe, in Chapters 9–11, wired and wireless serial links operating inside vehicles, in other words ‘internal’ systems (Safe-by-Wire Plus, 12C, D2B, CPL, MOST, IEEE 1394, etc.) and outside the vehicles, in other words ‘external’ systems (remote keyless entry, hands-free passive keyless entry, TPMS (tyre pressure monitoring system), tyre identification, GSM, Bluetooth, Zigbee and other related systems).

Acknowledgements

The field of multiplexed buses is constantly expanding, thanks to the work of many highly skilled people. I have been lucky enough to encounter many of them on numerous occasions, so it is very difficult for me to acknowledge all of them individually.

I must offer my special thanks to numerous friends at Philips Semiconductors of Nijmegen, The Netherlands, and Hamburg, Germany, with whom I have had the pleasure of working in this area for many years, and, at the risk of some unfairness, more particularly to Hannes Wolff, Matthias Muth, the many 'Hans' and other colleagues from the Netherlands, and the many 'Peters' and other colleagues in Germany.

Finally, I would be ungrateful if I did not also thank the many colleagues in the industry, including car makers and parts manufacturers, whom I regularly encounter at work meetings and at the ISO. They will know who they are from my remarks about the development of this book, as a result of which we all hope that this field of multiplexed buses will see the rapid growth that it deserves.

Lastly, I am extremely grateful to the Marcom team of Philips Semiconductors at Eindhoven, especially Manuella Philipsen, for the numerous documents and photographs which she was kind enough to supply and which so enliven this book and the cover.

Dominique PARET
Meudon, 10 june 2006.

Part A

CAN: From Concept to Reality

Let us be clear that we are talking about CAN (controller area network).¹ Yet another new protocol and system! True, but we must realize that it is not an easy matter to bring everybody into agreement, especially when everyone has his own objectives and specific technical requirements, and that major markets are sufficient to justify and optimize every concept (which clearly means reducing the cost!). So here is a new serial protocol to be added to the burgeoning family of local area networks (LANs).

As described by the ISO (International Organization for Standardization) standards, CAN is a ‘serial communication protocol which efficiently supports the distribution of commands in real time with a high security level. Its preferred fields generally relate to high bit rate, high transmission reliability network applications operating on a low-cost multiplexed cable principle’. You have been warned. . .

In this part of the book, I will describe the CAN protocol and some of its main applications.

I should point out that this concept was not developed overnight, but is the fruit of lengthy research and experimentation. Clearly, I could move straight on to a description of its operation, but I think this would deprive us of a depth of knowledge of the problems relating to buses used in local networks.

Indeed, some ideas commonly seem to spring up ‘naturally’, but often the major breakthroughs in our thinking are due to sequences of apparently unrelated events, which are in fact well organized.

So, before revealing the specifics of CAN as devised by R. Bosch GmbH, I wanted to provide an introductory chapter on the major features of this protocol, in an attempt to trace the thought processes followed by a large number of researchers in order to reach the final form of this project.

This approach is unusual, and although this introduction is not intended to be technical in nature, I advise you to read it carefully, as it will certainly give you a clearer understanding of all the ins and outs of the development of this very special protocol, which have made it so successful in the car industry and in other industrial and professional fields.

¹Sometimes ‘CAN bus’ is used, not out of ignorance, but simply to avoid possible confusion, which could easily cause an incorrect classification of the book, in web search engines for example. As my last word on this matter, the normal term in use is simply ‘CAN’, bearing in mind that the term ‘bus’ represents only one of the many possible applications topologies.

Note: To pay tribute where it is due, I have chosen to base the development of this introduction on a model similar to that used by one of the founders of the CAN bus, Professor Uwe Kiencke of the University of Karlsruhe, in an excellent presentation to the International CAN Conference, sponsored by CAN in Automation, in 1994 (Mainz, Germany) . . . proving that you can be a prophet in your own country!

1

The CAN Bus – general

A bus is always a bus – but there are ‘buses’ and ‘buses’!

In fact, from one bus to the next, the problems to be solved remain the same, but the different characteristics of the proposed fields of application modify the hierarchical order of the parameters to be considered, and result in the development of new concepts in order to find neat solutions to the difficulties encountered. Here is a quick list of the virtually unchanging problems encountered in bus and network applications:

- network access concepts, including, clearly, problems of conflict, arbitration and latency;
- deterministic or probabilistic aspects and their relationship with the concepts of real-time or event-triggered systems;
- the concept of network elasticity (‘scalability’);
- the security of the data carried, in other words, the strategy for managing errors including their detection, signalling and correction;
- questions of topology, length and bit rate;
- questions of physical media;
- radio-frequency pollution, etc.

1.1 Concepts of Bus Access and Arbitration

‘Distributed’ real-time control systems, based on an operating system located within a single processor, interconnected by a communication network with distributed processors, are currently providing a significant addition to ‘parallel’ systems.

In addition to the simple exchange of data, the processing must be synchronized, in other words, the execution must follow certain interrelated logical sequences. In these systems, the messages relating to synchronization are generally short. They can be created by any process or event in the system and must be simultaneously receiving, in order to maintain the coherence of parallel processing.

All stations independently generate messages concerning their respective tasks at random (event-triggered) instants.

The transmission requests contend with each other for access to the bus, leading to latencies¹ which are variable rather than constant.

Let us now examine the different principles of arbitration, which are in the running.

1.1.1 CSMA/CD versus CSMA/CA

CSMA/CD

For historic reasons, the Carrier Sensor Multiple Access/Collision Detect (CSMA/CD) arbitration procedure was initially developed to resolve these conflicts.

With this system, when several stations try to access the bus simultaneously when it is at rest, a contention message is detected. The transmission is then halted and all the stations withdraw from the network. After a certain period, different for each station, each station again tries to access the network.

It is well known that these data transfer cancellations during contention theoretically also decrease the carrying capacity of the network. The network may even be totally blocked at peak traffic times, which is unacceptable when the network is to be used for what are known as 'real-time' applications.

CSMA/CA

In view of the above problems, another principle (one of several) was carefully investigated. This is known as Carrier Sensor Multiple Access/Collision Avoidance (CSMA/CA).

This device operates with a contention procedure not at the time of the attempt to access the bus, but at the level of the bit itself (bitwise contention – conflict management within the duration of the bit). This principle has been adopted in the CAN (controller area network) protocol, and its operation is described in detail in this part of the book.

In this case, bus access conflicts can be avoided by assigning a level of priority to each message carried.

In the case of contention, the message having the highest priority will always gain access to the bus. Clearly, the latency of the messages will then depend markedly on the priority levels chosen and assigned to each of them.

1.1.2 The problem of latency

In the overall design of a system, in order to take all parameters into account, the latency is generally defined as the time between the instant indicating the request for transmission and the actual start of the action generated by this.

For the time being, and for the sake of simplicity, I shall define the latency of a message (t_{lat}) as the time elapsing between the instant indicating a request for transmission and the actual start of the transmission.

¹See Section 1.3.

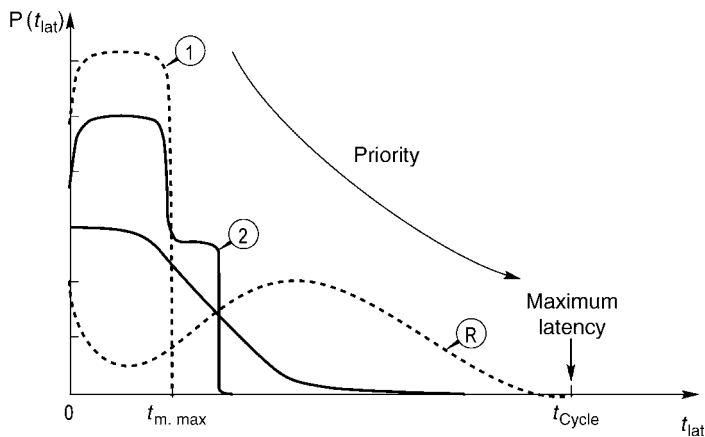


Figure 1.1

This concept is widespread and used in statistical analysis, mainly in what are known as ‘real-time’ systems. The reason is simple: only a few specific messages really need to have guaranteed latency, and then only during peak traffic times. We must therefore consider two kinds of messages:

- R = messages whose latency must be guaranteed,
- S = the rest,

and of course $M = R + S$, the total number of messages.

The curve in Figure 1.1 shows the probability distribution of the latency as a function of the latency, where the transmission request is made once only.

The specific value t_{cycle} is the time representing a mean activity cycle of the network consisting of M messages having a temporal length t containing N bits. The curves depend on the priority of the messages. The probability distribution of priority ‘1’ returns to 0 immediately after the transfer of the longest message.

1.1.3 Bitwise contention

This CSMA/CA principle, used in the CAN bus (patented since 1975), of the same type as that of the I2C bus, introduces some constraints, both in the representation of the signal in the physical layer and in relation to the maximum geometry ‘1’ of a network operating in this way.

Thus, during the arbitration phase, in order to have higher priority bits in the network erasing those of lower priority, the physical signal on the bus must be

- either dominant,¹ for example, the presence of power, current, light or electromagnetic radiation,
- or recessive,¹ for example, an absence of power.

¹Increase your Word Power.

By definition, when a dominant bit and a recessive bit are transmitted simultaneously on the bus, the resulting state on the bus must be the dominant state.

1.1.4 Initial consequences relating to the bit rate and the length of the network

Now, here are a few words about the consequences of what I have described above (a part of Chapter 3 will also deal with this thorny question).

We know that the propagation velocity of electromagnetic waves v_{prop} is of the order of $200,000 \text{ km s}^{-1}$ in wires and optical fibres (or, expressed another way, each wave takes approximately 5 ns to cover 1 m , or again travels at $200 \text{ m } \mu\text{s}^{-1}$).

Theoretically, in a system operating by bit contention, a bit can travel from one end of the network to the other before being detected on its arrival.

Now, it is possible that, a few ‘micro-instants’ before the bit reaches its destination, the other station, having seen nothing arrive at its terminal, may decide to start transmitting in the other direction. Head-on collision! Disaster looms!

If we call t_{bus} the time taken by the signal to travel the maximum length of the network, the global sum of the outward and return times for the propagation of the signals on the bus is

$$2t_{\text{bus}} = 2 \frac{l}{v_{\text{prop}}}.$$

For example, if $l = 40 \text{ m}$, then $t_{\text{bus}} = 200 \text{ ns}$.

To enable the station which sent the initial bit to manage the conflicts, the necessary duration of the bit, known as the bit time or t_{bit} , must be longer than t_{bus} . And for the sake of completeness, we must allow for the time taken (or required) to sample and process the bit at the station where it arrives.

To evaluate the minimum bit time, $t_{\text{bit-min}}$, of the proposed network, it is necessary (Figure 1.2) to allow for

- the outward propagation delays, t_{out} ,
- the inward propagation delays, t_{in} ,

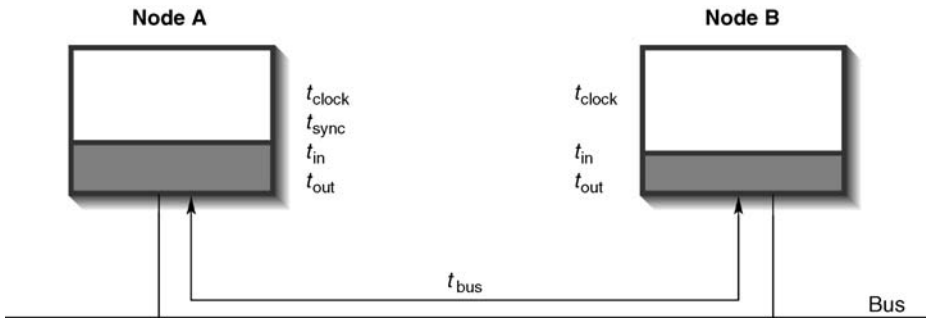


Figure 1.2

- the delays due to synchronization, t_{sync} ,
- the phase differences due to clock tolerances, t_{clock} ,

giving a total $t_{\text{bit-min}}$ of

$$t_{\text{bit}} = 2t_{\text{bus}} + 2t_{\text{out}} + 2t_{\text{in}} + t_{\text{sync}} + t_{\text{clock}}.$$

Example.

With a bit rate of 100 kbit s^{-1} , that is a bit time of $10 \mu\text{s}$, we can achieve a network length of approximately 900 m.

All these concepts are described in detail in Chapter 3.

1.1.5 The concept of elasticity of a system

The architectural and topological configuration of a distributed system is generally different from one application to another, and, for a single application, it can also develop or be changed over a period, depending on the requirements to be met. To make things clearer, consider the example of the installation of industrial production lines, which, although they always have the same overall appearance, need to be reconfigured from time to time according to the products to be made. Where systems or networks are concerned, we generally use the term ‘elasticity’ to denote the capacity to withstand a change of configuration with the least possible amount of reprogramming in relation to the data transfer to be provided.

Let us look more closely at the problems associated with the elasticity of a network.

The information received and processed somewhere in a distributed system must be created and transmitted to a station. There is no logical alternative to this. There are two possible cases:

- New information is to be added. Any new information requires a new message transfer and therefore a reprogramming of the communication. In this case, the station which previously sent this specific information element must be reprogrammed according to the new one, whereas the other stations remain unchanged.
- A different situation occurs when a pre-existing specific information element has to be either transmitted from another station or received by additional stations. In this case, the additional receiving station must be reprogrammed to receive the information.

1.1.6 Implication of the elasticity of a system for the choice of addressing principle

Conventional addressing, generally consisting of a ‘source’ address on the one hand and a ‘destination’ address on the other hand, cannot provide a system with good structural elasticity. This is because any messages that have to be rerouted will require modifications, even if this is not logically necessary, as mentioned above.

For the CAN concept, in order to provide good elasticity in the system, it was decided to use another addressing principle, based not on the source and destination addresses but on the content of the message itself. This implies two things:

- A message has to be transmitted to all the other stations in the network (the term used for this principle is ‘broadcast diffusion’).

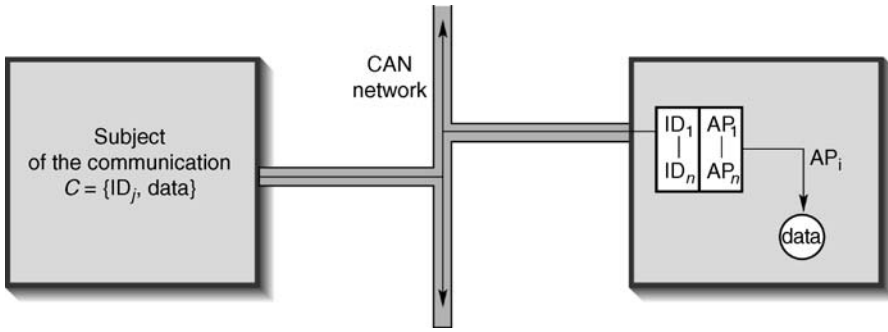


Figure 1.3

- The selection processing of the transmitted message is then carried out by what is called ‘acceptance filtering’ at each station.

For this purpose, the message is labelled with an identifier $ID(i)$, which is then compared with the list of messages received (or to be received) at each station.

This list contains address pointers $AP(i)$ towards the communication buffer, enabling the content of the message to be stored (Figure 1.3).

Therefore, *all* the messages are simultaneously received over *all* of the network, and the data consistency is thus guaranteed in distributed control systems.

1.2 Error Processing and Management

1.2.1 The concept of positive and negative acknowledgements

The elasticity of systems and identification based on the message content complicate the processing of errors when they occur.

An example of a conventional method of (non-)detection of errors is the return of what is called a ‘positive’ acknowledgement from the receiving station to the transmitting station, when a message is received correctly. The key information contained within such a positive acknowledgement is generally the address of the receiving station.

In the CAN concept, this idea of a local address completely disappears, and the identifier ‘labelling’ the message is transmitted to all the participants and received everywhere in the network. This makes it necessary to execute a local error processing task in each of the stations in the network, in the absence of such local message addresses. To achieve this, the CAN protocol concept uses a combination of positive and negative acknowledgements.

The positive acknowledgement $ACK+$ is expressed as follows:

$$ACK+ = ACK + (i) \text{ for any } (i).$$

It is sent from all stations (i) which have received the message correctly and expresses this (positive) acknowledgement during a single precisely defined time interval (ACK time slot).

The principle described above can also be expressed in two basic forms:

- an optimistic view of the problem, in which we can say that the positive acknowledgement indicates that ‘at least one station has received the transmitted message correctly’;
- a rather less optimistic (but not totally pessimistic) view: ‘The negative acknowledgement of a message must take a form such that it indicates (or will indicate) that there is at least one error in the global system’.

In the latter case, the message indicating the presence of the error must be sent over the network immediately after the detection of the error. This method will ensure that the system can be resynchronized immediately within a single message frame. The latter point is crucial for the security of the applications considered.

1.2.2 Error management

The combination, the redundancy and the analysis of the positive and negative acknowledgements sent from the error processing devices of the different stations are exploited to provide strong presumptions concerning the source of an error (either from the transmitting station or from one of the receiving stations), for example

- The presence of at least one positive acknowledgement sent from a receiver, combined with an error message, signifies that the message has been transmitted correctly at least.
- Conversely, the absence of a positive acknowledgement, together with an error message, indicates that all the receiving stations have detected an error and that there is a strong presumption (or probability) that the error is located in the transmitting station.

1.2.3 Error messages

The error messages used for the CAN concept are of two types (Figure 1.4).

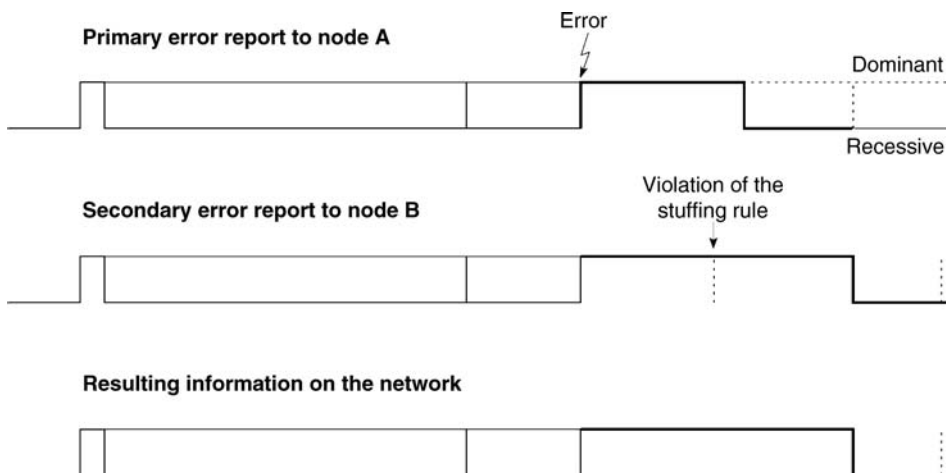


Figure 1.4

Primary error report

A station (or several stations at once) detect(s) an error, causing it (or them) to immediately transmit an error message.

At this point, the error message consists of a sequence of dominant bits and the current message is aborted (not fully processed).

Secondary error report

Following this cancellation of the message, all the other stations detecting a violation of the format rules transmit an error message on their own behalf, these messages being combined with the first one and thus increasing its duration.

Primary error reports are more probable for a station where the error has occurred than secondary error reports, which are generated in response to the aforementioned error.

1.2.4 The concept of an error management strategy

To manage errors correctly, the ultimate aim is to develop a strategy for the processing of errors. The quality of the processing will depend markedly on whether or not the strategy defined for a given field of applications is proactive.

For the CAN protocol, a strategy called an error (or fault) ‘confinement’ strategy, shown in outline in graphic form in Figures 1.5 and 1.6, has been defined (and is described in much greater detail in Chapter 2). Here is a brief summary.

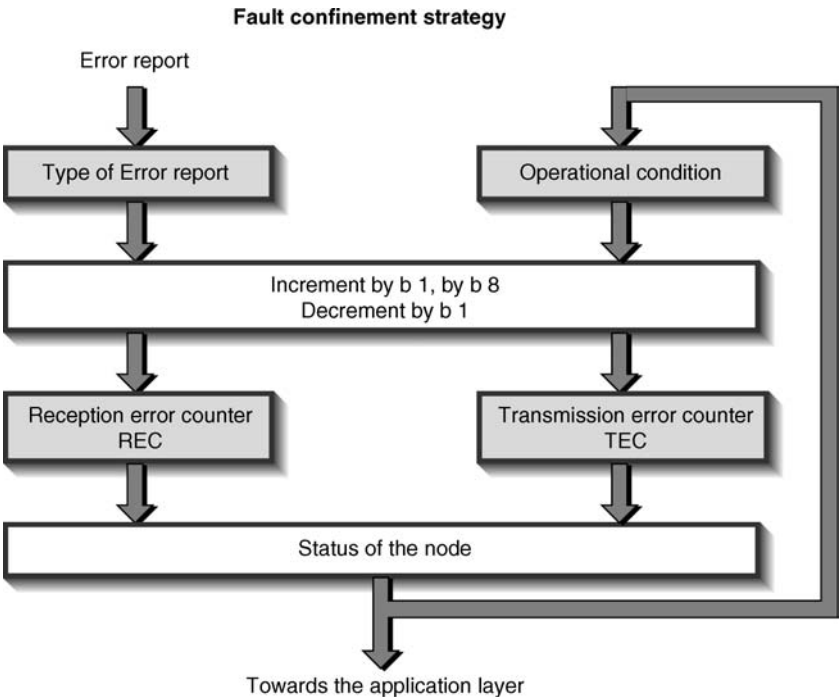
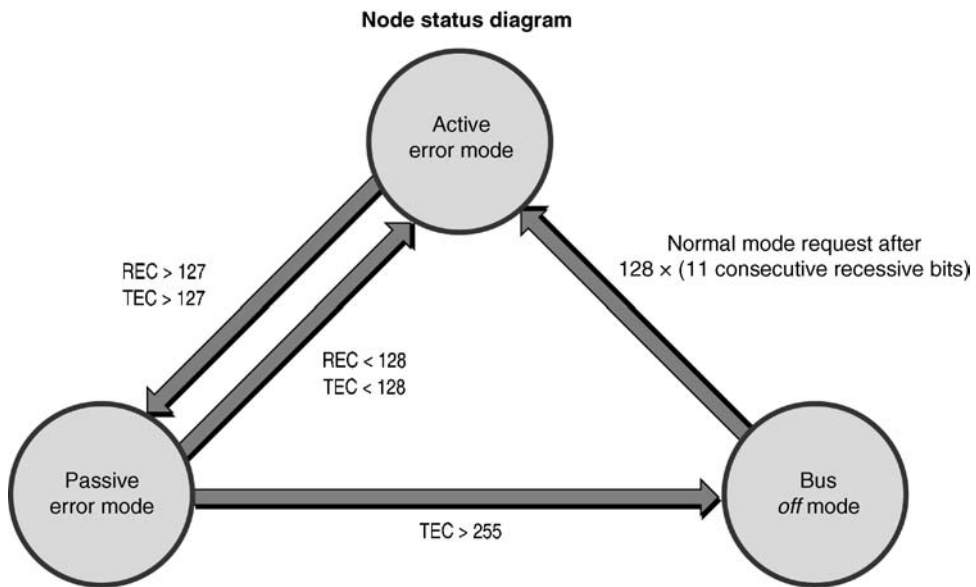


Figure 1.5

**Figure 1.6**

First, each station has to have two separate error counters: one to record what happens during the transmission of a message and the other to execute a similar task during reception.

Then, according to the type of error report and the operating conditions of the station at the instant in question, the counters are incremented with different weightings according to certain conditions or are decremented.

The purpose of these counters is to record information originating directly or indirectly from all the other stations. Using this device, the counters carry out an averaging operation, giving an approximation of the statistical values of the local quality of the network at a given instant.

To provide more support for this strategy, it was also specified that if too many errors were attributed by such statistical means to a specific station, its state would be transferred from what is called an ‘active error’ mode to a ‘passive error’ state, in which it would no longer communicate but would always be active in the management of errors which may occur on the network.

If there are too many transmission errors in the network, the network could be blocked, making all traffic impossible.

To avoid this, it is necessary to specify that, above a certain limit (fixed at 255 for CAN), the station switches to another new state, called *bus off*, during which the station appears to be physically disconnected from the bus so as not to block the bus for any reason.

In this case, its input stage stays in a high impedance state, thus making it possible to observe the signal travelling in the network (monitor function).

This ends the summary of the key points for the study of this new protocol concept, together with the major ideas used to resolve the problems generally raised by the applications.

At this stage, you have the necessary background to study the actual operation of this protocol and of the associated bus.

Now for the work! The bus is waiting, so let us go. . . (see also *The I2C Bus* [same author, same publisher] in which, at that time, we were only waiting for the. . . bus!).

1.3 Increase Your Word Power

To ensure that we are using the same vocabulary, I now offer for your guidance a few unillustrated extracts from dictionaries:

- *Avoidance*: the fact of avoiding, from the verb ‘to avoid’.
- *Confinement*: the act of confining (keeping within confines, limits, edges).
- *Consistency*: keeping together, solidity.
- *Contention*: argument, dispute, from the Latin *contentio* (struggle, fight).
- *Identifier*: ‘that which identifies’.
- *Latent*: in a state representing latency (see ‘latency’).
- *Latency*: time elapsing between a stimulus and the reaction to the stimulus.
- *Recessive*: persisting (still active) in the latent state.

1.4 From Concept to Reality

The foundations of the CAN concept were laid, at a given period, on the basis of a state of the market and technology, with an attempt to extrapolate future trends, in respect of the systems and the techniques and technologies to be implemented to make them a reality.

The question that you may well ask is: Why CAN and not another protocol?

So here are some further explanations which will help you to understand; let us begin with a few words about ‘site buses’.

1.4.1 The site bus market

For a long time, many companies have been obliged to develop and suggest their own solutions for resolving substantially similar (or related) problems raised by links and communications between systems.

Nearly all of these solutions are known as ‘proprietary’, and because of the different interests of each of the companies involved, this has led to a significant fragmentation of the market. This diversification of the solutions and the small or medium quantities covered by each of them have therefore, unfortunately, led to a decrease in the quantities of specific components to be developed and produced in order to create a standard on this basis. Among these solutions, many serial buses and links have existed for a long time to resolve similar or virtually similar applications. Regardless of the markets supported (industrial or motor vehicle), the best known names (listed in alphabetical order to avoid offending certain sensitivities) are Batibus, Bitbus, EIB, FIP bus, J1850, LONwork, Profibus, VAN, etc.

Figure 1.7 provides a summary of the numerous buses for covering some or all of these applications.

So why use CAN, which is also a proprietary system?

1.4.2 Introduction to CAN

The size of the motor vehicle market (several million units per year) has led component manufacturers to design integrated circuits handling the CAN protocol. It has also

	LON	CAN	FIP	Profibus	Bitbus	Interbus
General						
Name of protocol	Local operating network	Controller area network	Factory instrumentation protocol	Process field bus	Bitbus	Interbus
Fields of application	Industrial domestic automation	Motor vehicles, industrial	Industrial	Industrial	Industrial	Industrial
Original designer	Echelon Corp.	R. Bosch	Schneider	Profibus Consortium	Intel	Phenix contact
Country	United States	Germany	France	Germany	United States	Germany
Users	LONUSERS	CIA – CAN in Automation	FIP-Club International	Profibus Nutzer Organization		
Specific characteristics of the physical layers						
Topology	Bus	Bus	Bus	Bus	Bus	Ring
Media supported	Diff. pairs RS 485, CPL infrared	Diff. pairs RS 485, Infrared CPL Optical fibres	Diff. pairs RS 485	RS 485	RS 485	RS 485
Max. length (m)	500	40		1200	1200	
For a bit rate (Mb) of	1.25	1				
General characteristics of the protocols						
ISO/OSI layers	2 to 7	1 and 2	2, 7	1, 2, 7	1, 2, (5), 7	1, 2, 7
Min. bit rate (kbs)	4.9	1000	31.25	9.6	62.5	300
Max. bit rate (kbs)	1250		1000	500	2400	300
Max. no. of bytes in message	249 (+7)	8(+2)		246 (+2)	128 (+2)	
Priorities	128	2048		2		
Application layers	Lontalk	Numerous	FIP	FMS	RAC	PMS
Electronic components (stand-alone or microcontroller) for protocol handlers						
References	3150, 3120	^a			8 × 44, 80C 152	IPMS
Manufacturers	Motorola, Toshiba	^a		Siemens	Intel	

^a The list is too long to be shown here.

Figure 1.7

succeeded in reducing costs significantly, which is not necessarily the case with other buses, which are frequently restricted to smaller scale applications without the benefit of dedicated components or keen prices. Manufacturers have therefore considered the arrival of the CAN bus from another viewpoint (the very special viewpoint of the performance/cost ratio) and have suddenly decided that this type of bus fully satisfies them.

It is surprising what can be achieved by taking a fresh look!